Databases and corpora 1

Quantitative Methods in Historical Linguistics

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First things first

- Any questions?
- N.B. `hdist.R` available on ILIAS as an alternative to the `rdist` package
- Course evaluation
- Download `ellegard.csv` from ILIAS (folder 'Data') – we will need this later today

A small thing: NA in R

- In R, `NA` is a special value meaning "not available"
- `NA` values have an effect on some commands, e.g. `mean` and `median`:

```
x <- c(1,5,3,NA,2)
mean(x)

## [1] NA

median(x)

## [1] NA
```

- Mean and median of `x` are undefined because of the `NA`

A small thing: NA in R

- To fix this, we can instruct `mean` and `median` to disregard the `NA` value(s)
- The relevant optional argument is `na.rm`:

```
x <- c(1,5,3,NA,2)
mean(x, na.rm=TRUE)

## [1] 2.75

median(x, na.rm=TRUE)

## [1] 2.5
```

Another small thing: tally

- Who are still with us?

Image credits: https://commons.wikimedia.org/wiki/File:Forex_-_14600958045.jpg

Language (community) vs. stock market

- What do language communities and stock markets have in common?
- Both are communities of people who interact (**AGENTS**)
- The agents' interactions lead to **EMERGENT** properties of the entire system
- These properties may be observed and described
- E.g. the fluctuations in the value of a stock over time
- E.g. the frequency of use of a certain linguistic variable over time

Change of perspective

- Up to this point we've been looking at phylogenetic trees
- In such a tree, every branching point (internal node) corresponds to at least one linguistic change
- Now we're "zooming in" to these branching points to see what individual changes look like

Databases and corpora

- A **CORPUS** (pl. **CORPORA**) is a collection of texts, usually in one language
- A **DIACHRONIC** corpus contains texts from many different time points (a non-diachronic corpus is a **SYNCHRONIC** corpus)
- A linguistic **DATABASE** is a collection (diachronic/synchronic) of data from a language
- A database can be produced from a corpus by looking at certain variables of interest in the corpus
- In this course, our interest is mainly in databases — to learn more about corpora, see a course/book in corpus linguistics

Example: English *do*-support

- In Modern English, questions require a form of the auxiliary *do* (known as **DO-SUPPORT**):
  (1) Do you like tea?
  (2) *Like you tea?
- Not so in Middle English:
  (3) How great and greuous tribulations suffered the Holy Appostyls?
- So there must have been a change: no do-support > do-support
- It is then interesting to ask: if we tried to draw a curve that gives the frequency of *do*-support as a function of time, what would that curve look like?

⇒ We need to collect a diachronic corpus of texts and turn it into a database that gives the required frequencies

Example: English *do*-support

- Luckily for us, Alvar Ellegård already did this in the 1950s
- So let's use his data
- But first, what do we mean by "frequency"?
- **Raw frequency** of x = number of times x appears
- **Relative frequency** of x = proportion of times x appears (rather than not-x)
- E.g. relative frequency of *do*-support is calculated as

$$\frac{\text{\# sentences with } do\text{-support}}{\text{\# all sentences}}$$

- Relative frequencies are always bounded between 0 and 1

Small print: if I say "frequency", I probably mean relative frequency. If I mean raw frequency, I'll (try to remember to) say "raw frequency".

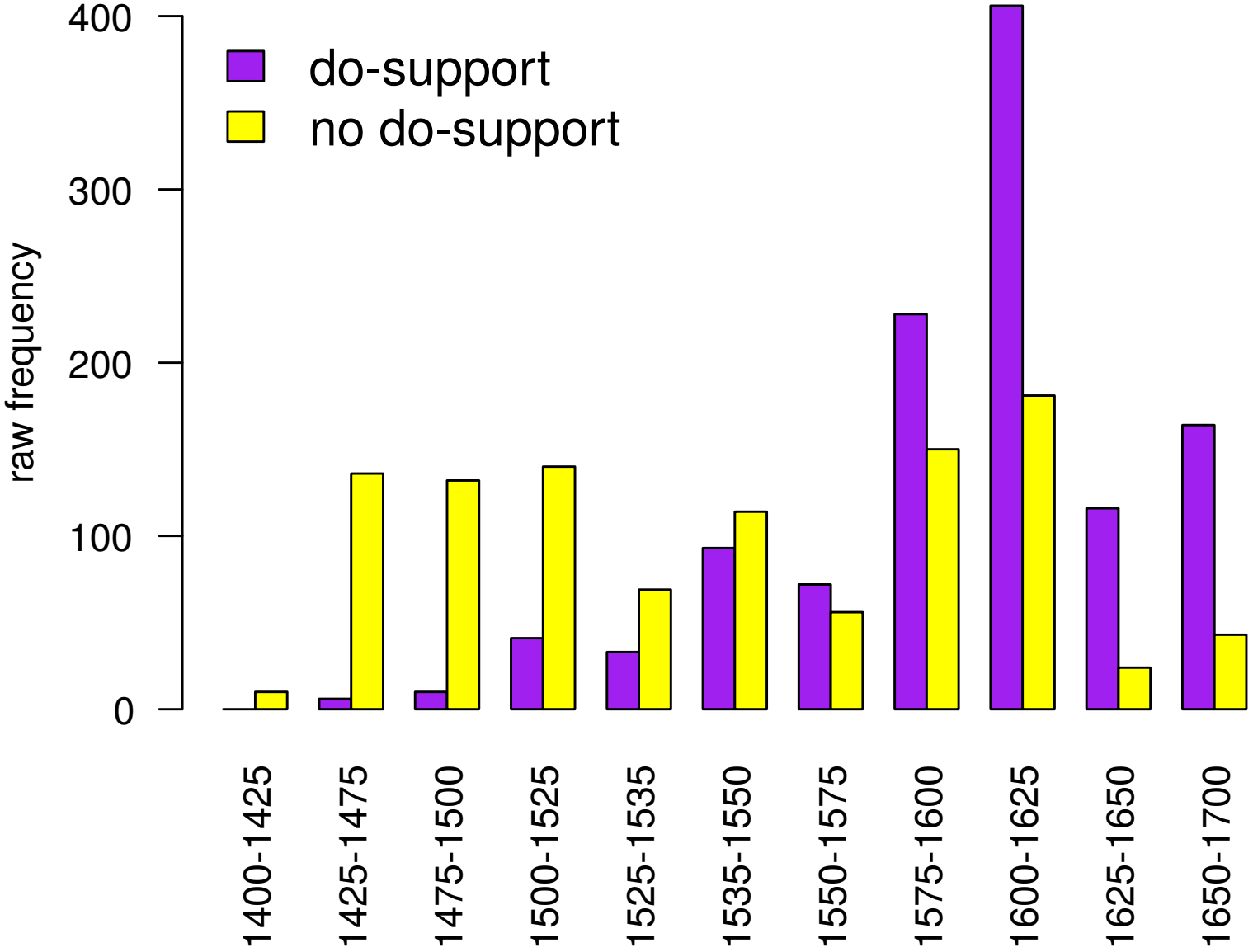
Example: English *do*-support

- Here's (some of) Ellegård's[1] raw data from his corpus:

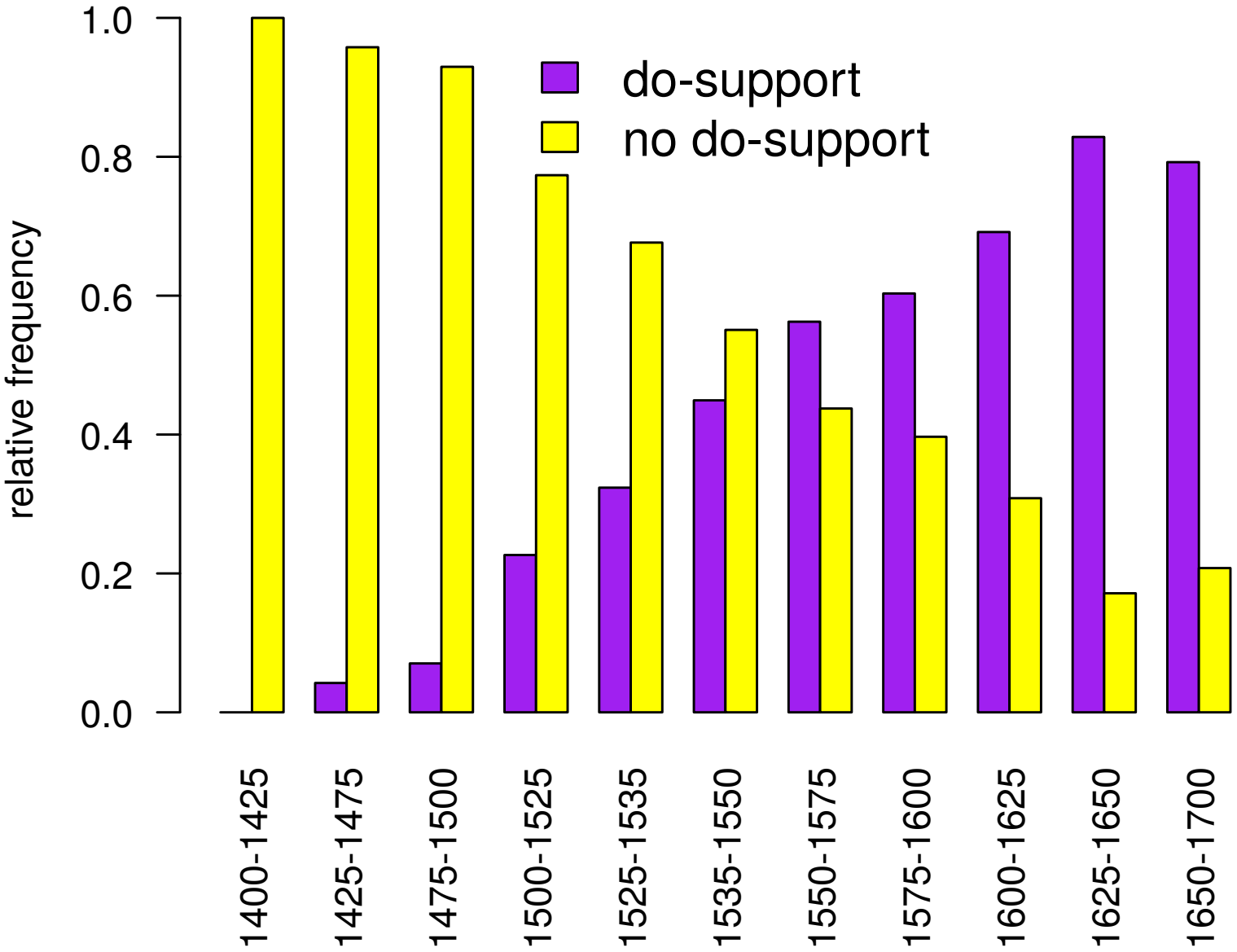
	questions (= interrogative sentences)	
time period	with *do*-support	without *do*-support
1400–1425	0	10
1425–1475	6	136
1475–1500	10	132
1500–1525	41	140
1525–1535	33	69
1535–1550	93	114
1550–1575	72	56
1575–1600	228	150
1600–1625	406	181
1625–1650	116	24
1650–1700	164	43

[1] Ellegård, A. (1953) *The auxiliary do: the establishment and regulation of its use in English*. Stockholm: Almqvist & Wiksell.

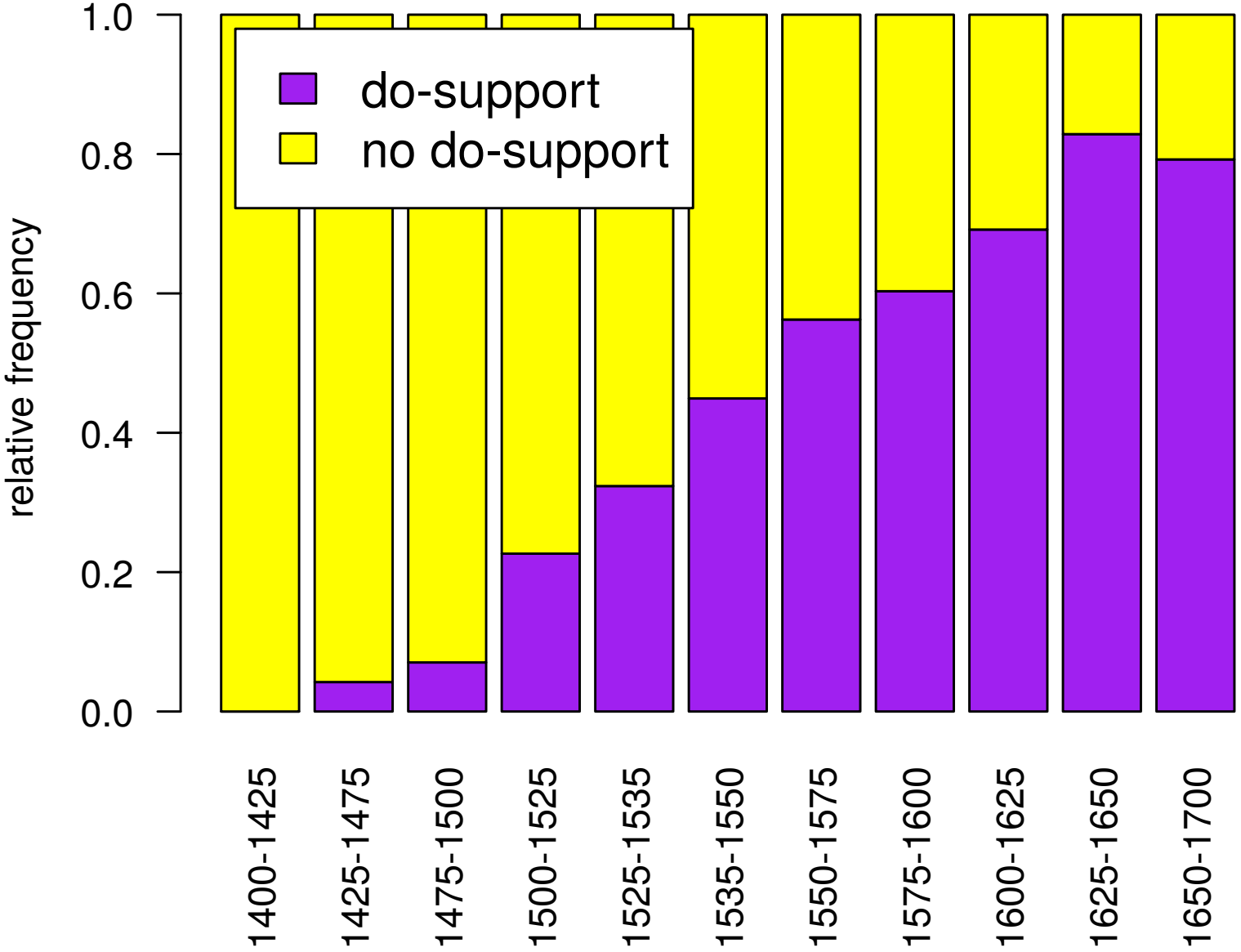
Example: English *do*-support



Example: English *do*-support



Example: English *do*-support



Example: English *do*-support

- Now let's try this for ourselves:

```
elleg <- read.csv("ellegard.csv")
head(elleg)
barplot(elleg$AQ_do, names.arg=elleg$period)
```

- Add some fancy stuff:

```
barplot(elleg$AQ_do, names.arg=elleg$period, las=2)
barplot(elleg$AQ_do, names.arg=elleg$period, las=2,
        col=c("purple", "yellow"))
barplot(elleg$AQ_do, names.arg=elleg$period, las=2,
        col=c("purple", "yellow"), ylab="raw frequency")
```

Example: English *do*-support

- How to plot both frequencies?
- We need to `rbind` the relevant columns from the data frame:

```
both_freqs <- rbind(elleg$AQ_do, elleg$AQ_nodo)
barplot(both_freqs, names.arg=elleg$period, las=2)
barplot(both_freqs, names.arg=elleg$period, las=2,
        beside=TRUE)
```

- Barplots can be plotted horizontally, too:

```
barplot(both_freqs, names.arg=elleg$period, las=2,
        beside=TRUE, horiz=TRUE)
```

Example: English *do*-support

- How do we get *relative* frequencies?

Example: English *do*-support

- How do we get *relative* frequencies?
- Divide by the sum of both raw frequencies:

```
rel_do <- elleg$AQ_do/(elleg$AQ_do + elleg$AQ_nodo)
rel_nodo <- elleg$AQ_nodo/(elleg$AQ_do + elleg$AQ_nodo)
```

- Now we can plot:

```
barplot(rel_do, names.arg=elleg$period, las=2)
barplot(rbind(rel_do, rel_nodo), names.arg=elleg$period,
        las=2)
```

The time dimension

- We have plotted the adoption of *do*-support in (Early Modern) English using Ellegård's time periods
- Problem: the periods are not all of the same length
  - 1500–1525 ⇒ 25 years
  - 1525–1535 ⇒ 10 years
  - 1535–1550 ⇒ 15 years
- One solution: use time period midpoints

Reflection Exercise

If you know the start and end years of a time period, how to calculate the period's midpoint?

The time dimension

- We have plotted the adoption of *do*-support in (Early Modern) English using Ellegård's time periods
- Problem: the periods are not all of the same length
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Reflection Exercise

If you know the start and end years of a time period, how to calculate the period's midpoint?

- Answer: If the period starts at S and ends at E, the midpoint occurs at $S + (E - S)/2$

The time dimension

- The `start` and `end` columns of the Ellegård dataset contain the period start and end years
- From these, we can easily calculate the period midpoints:

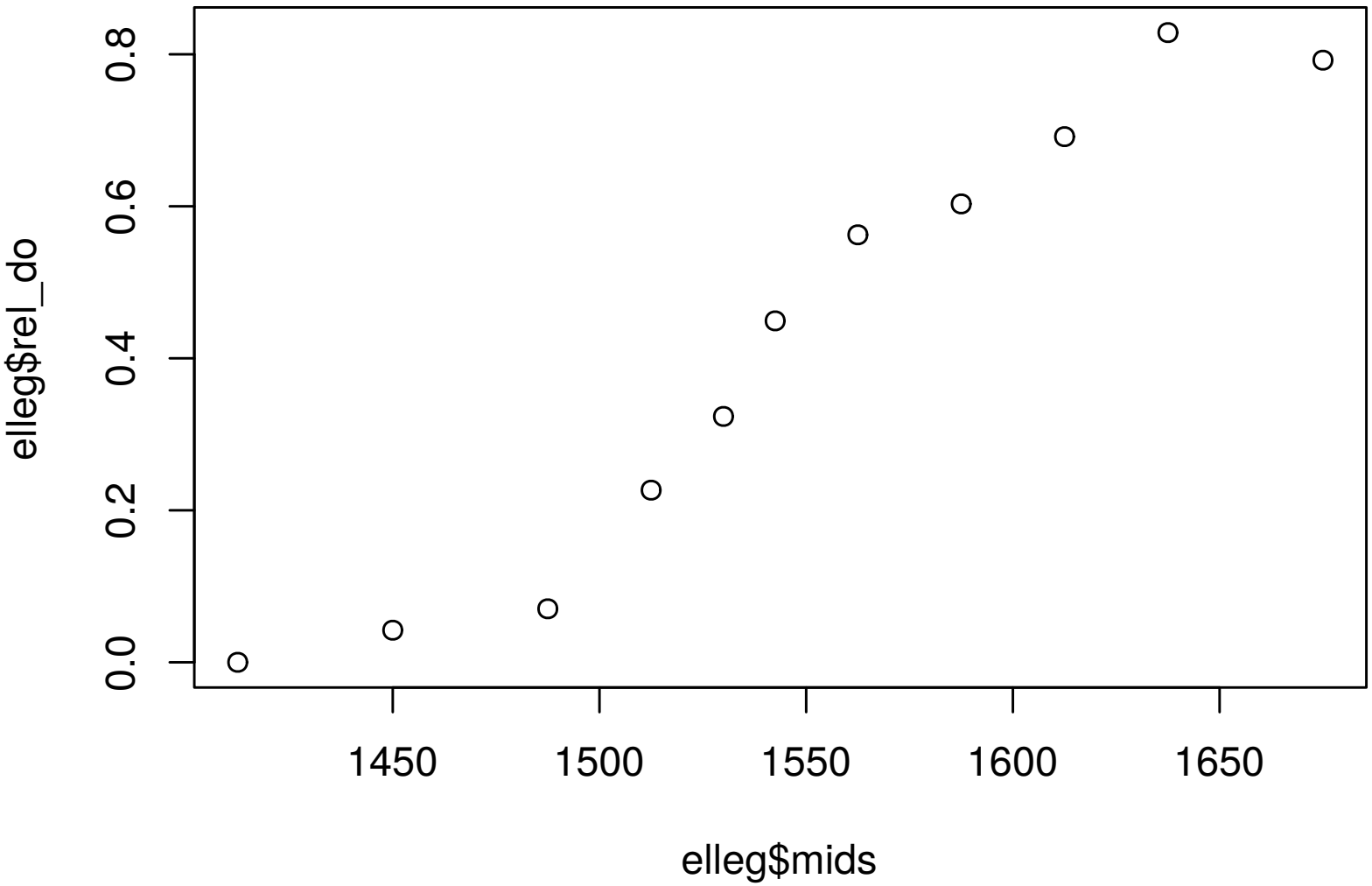
```
elleg$start + (elleg$end - elleg$start)/2
```

- We can even add these midpoints as a new column of `elleg`:

```
elleg$mids <- elleg$start + (elleg$end - elleg$start)/2
```

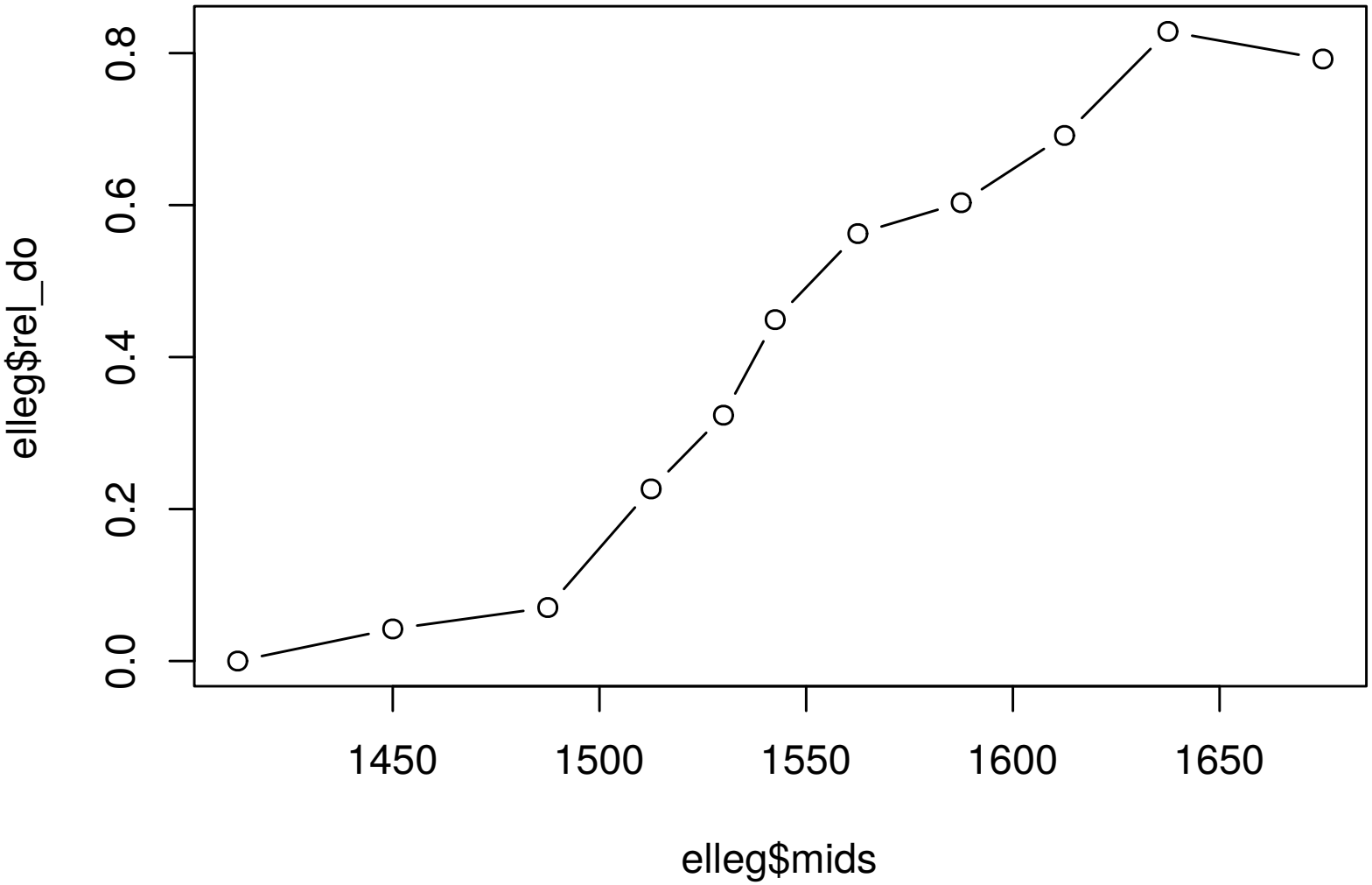
The time dimension

```
elleg$rel_do <- elleg$AQ_do/(elleg$AQ_do + elleg$AQ_nodo)
plot(elleg$mids, elleg$rel_do)
```



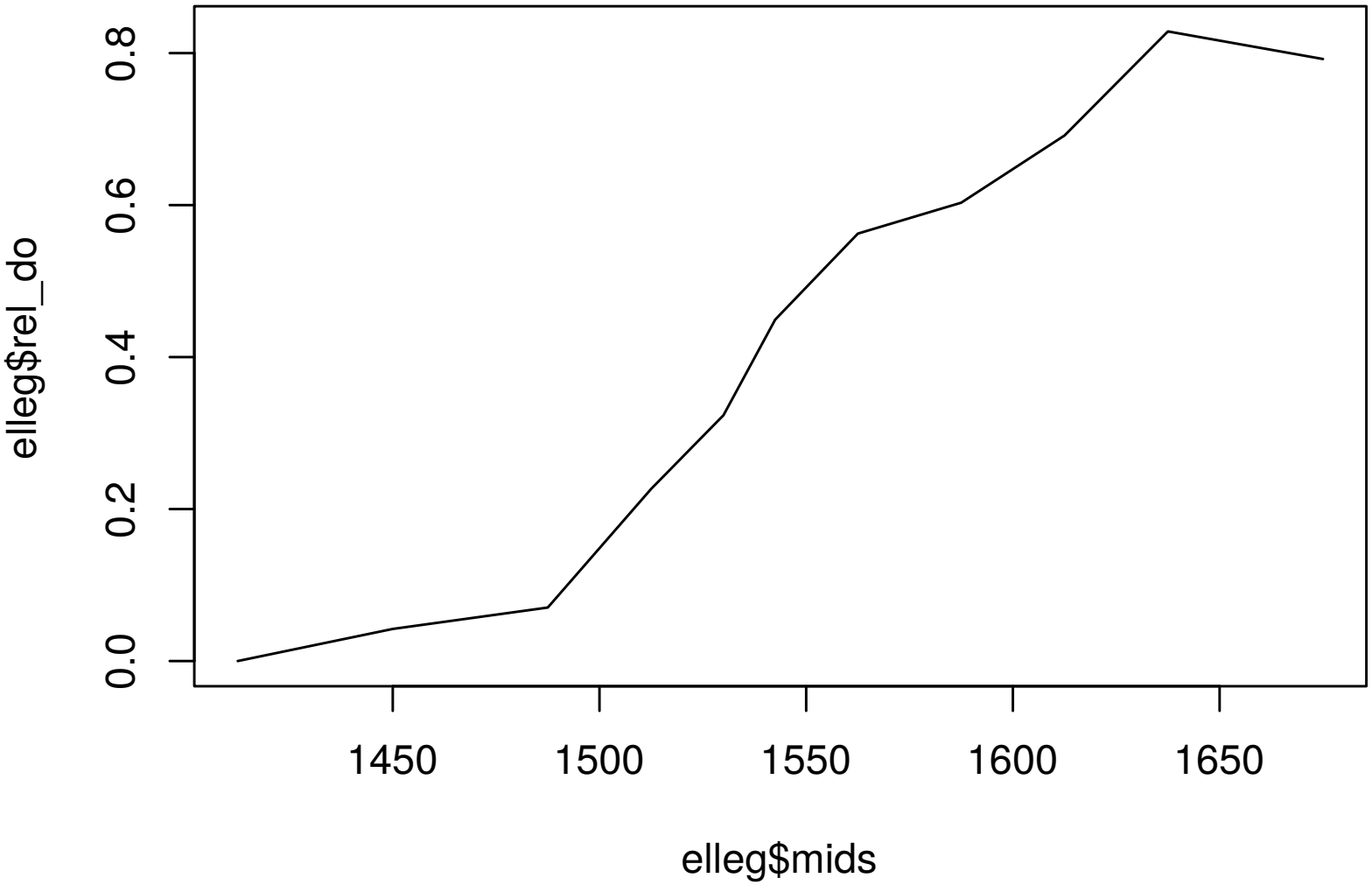
The time dimension

```
plot(elleg$mids, elleg$rel_do, type="b")
```

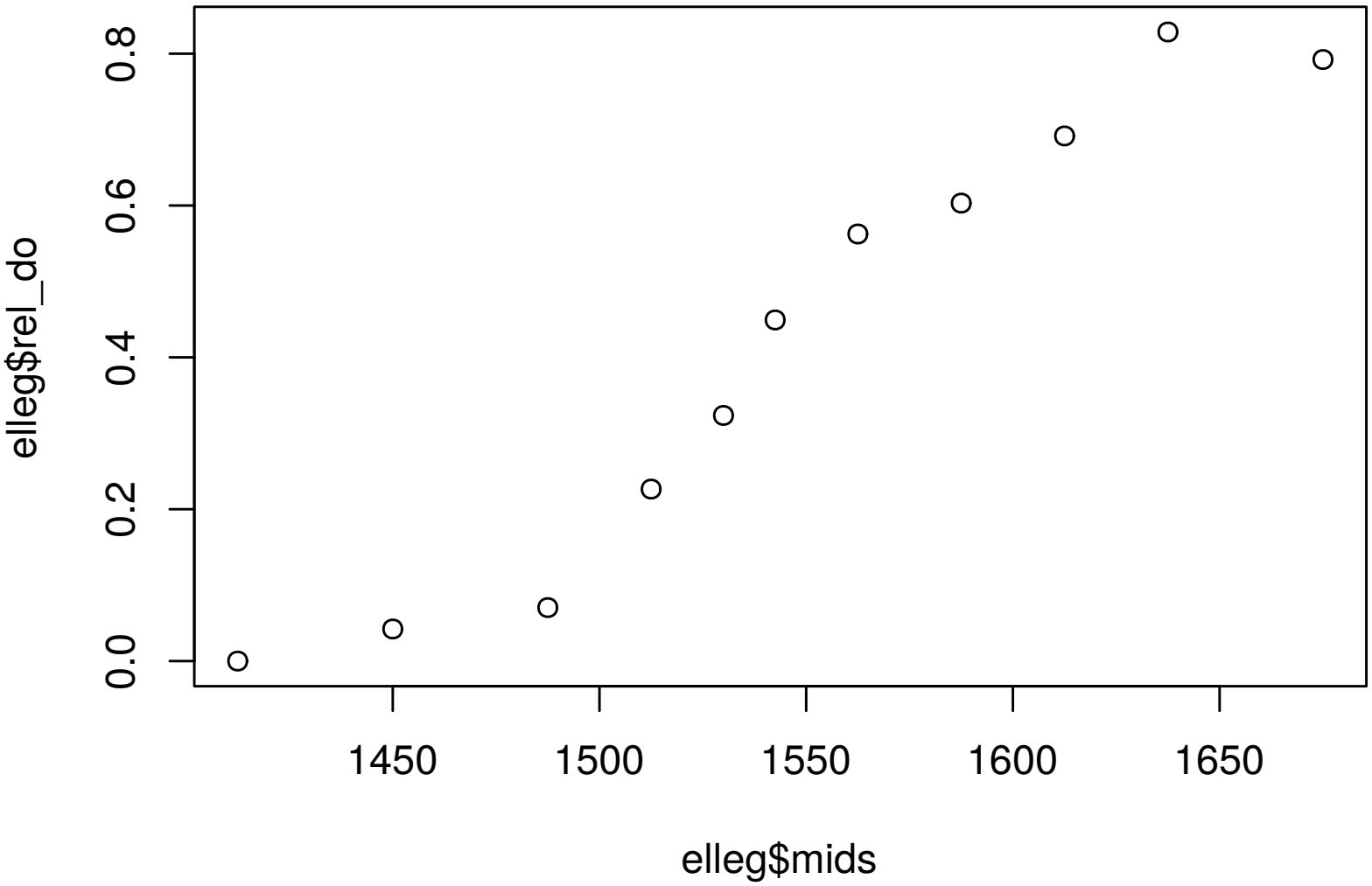


The time dimension

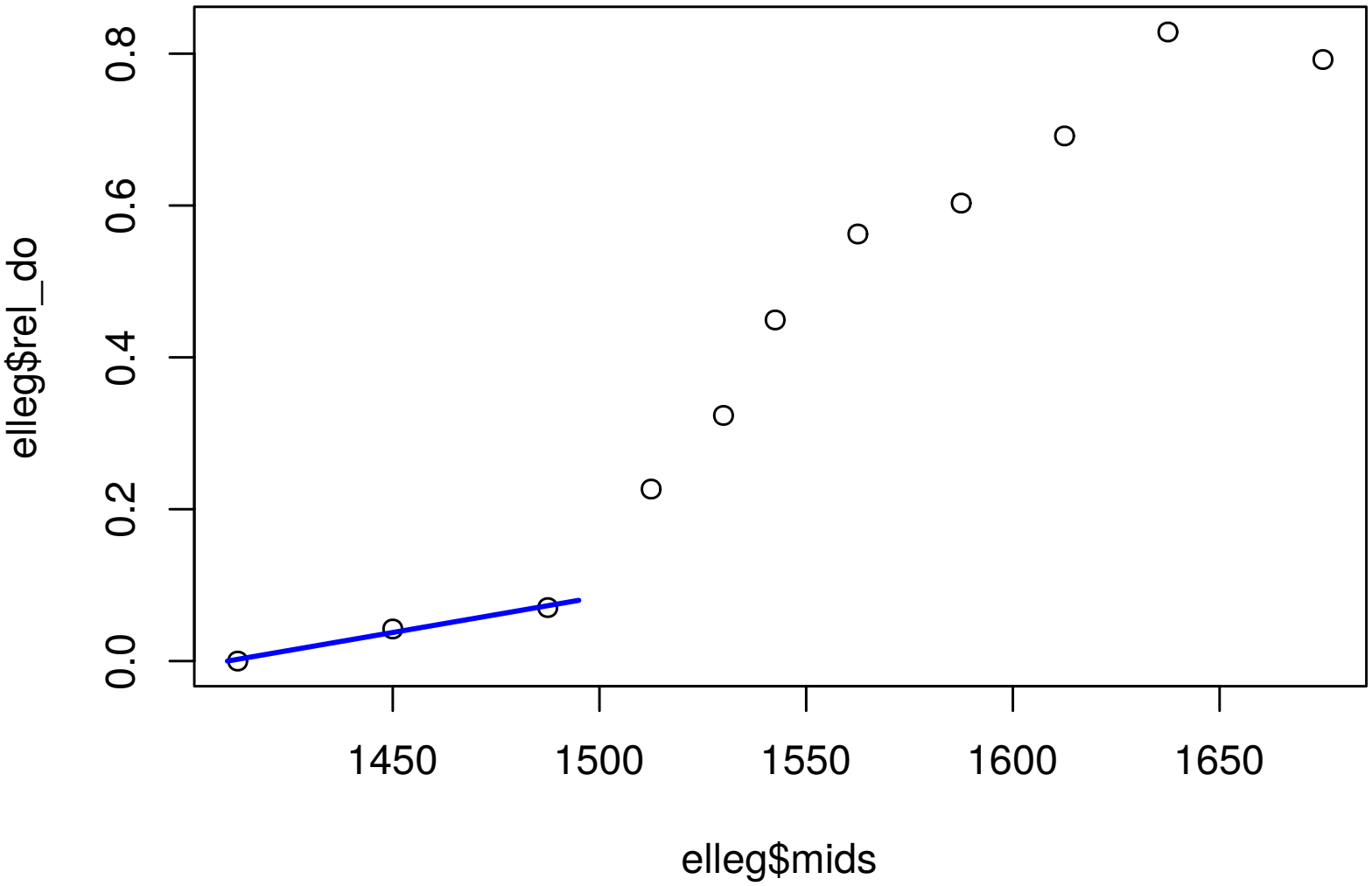
```
plot(elleg$mids, elleg$rel_do, type="l")
```



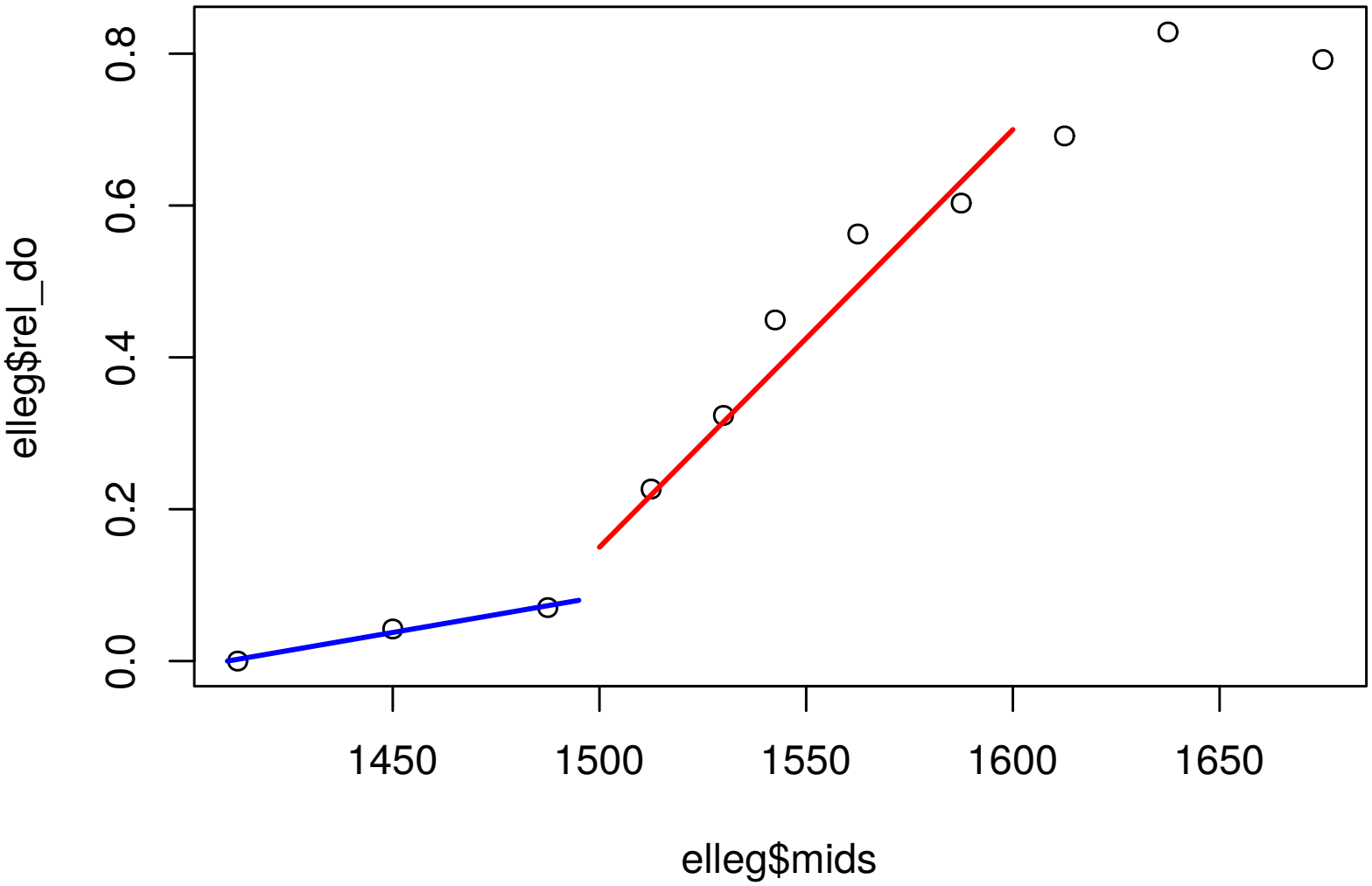
The time dimension



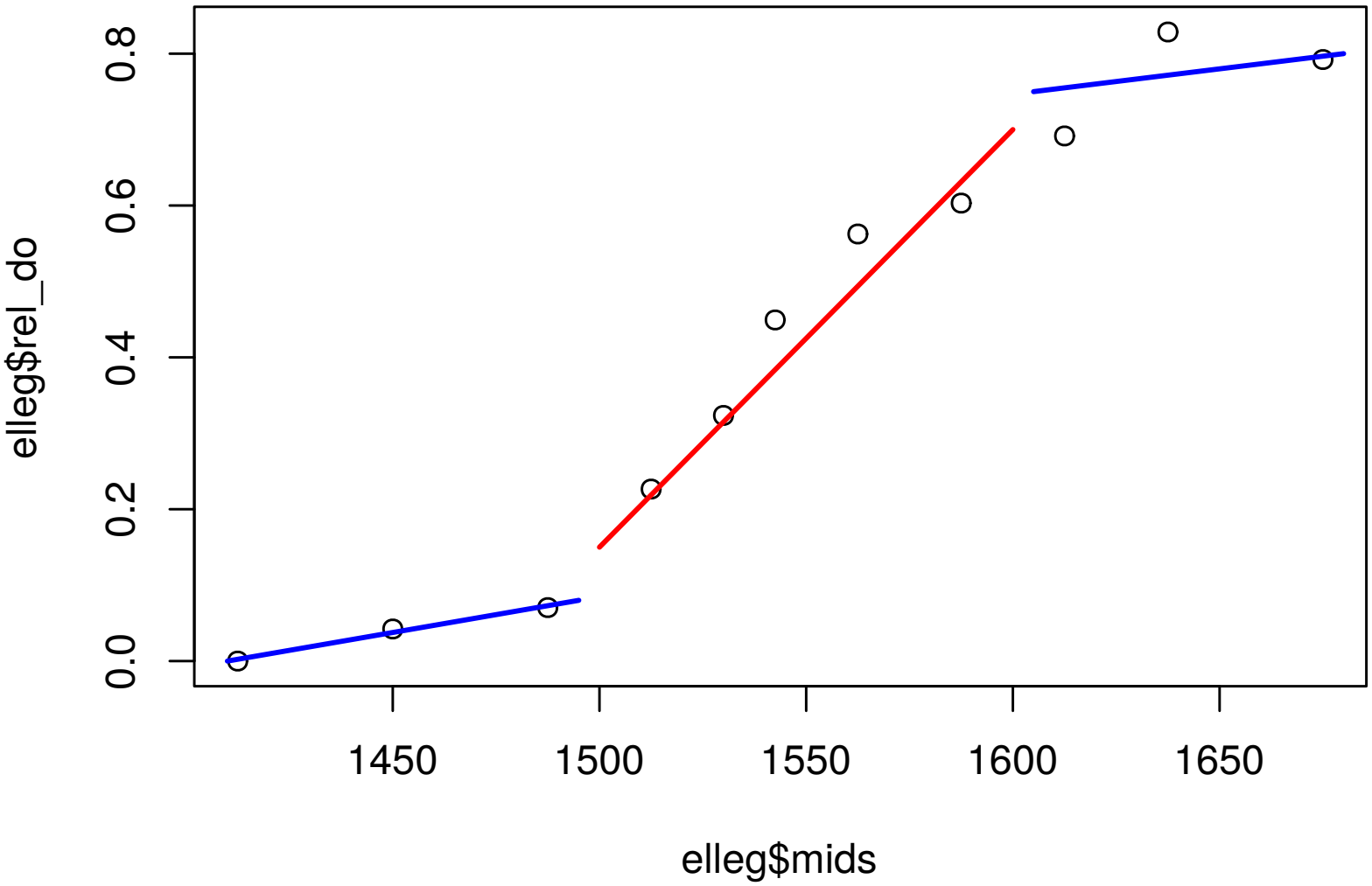
The time dimension



The time dimension



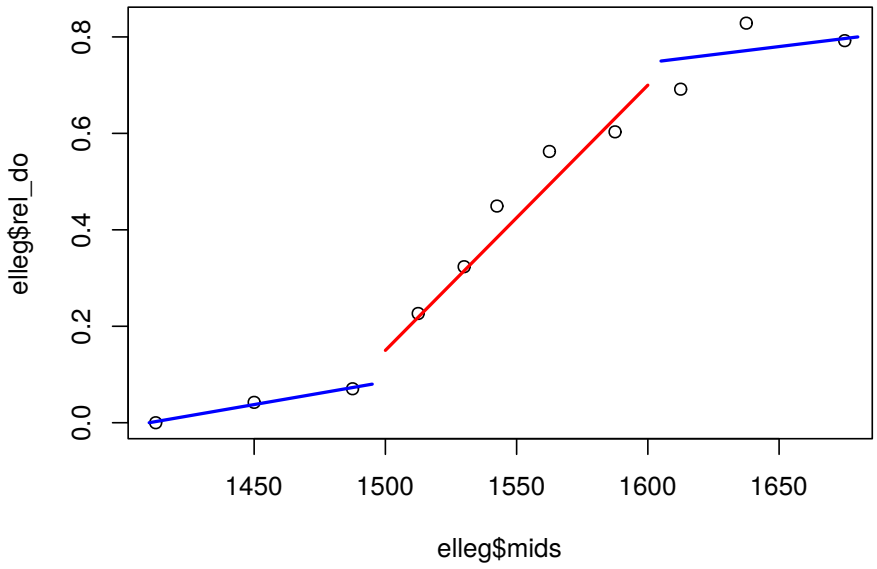
The time dimension



The time dimension

- The change starts slow, then speeds up, then slows down again
- A few questions:
  (1) Why does it do this?
  (2) Do other changes have this “slow–quick–slow” pattern?
  (3) How can the pattern be precisely described?

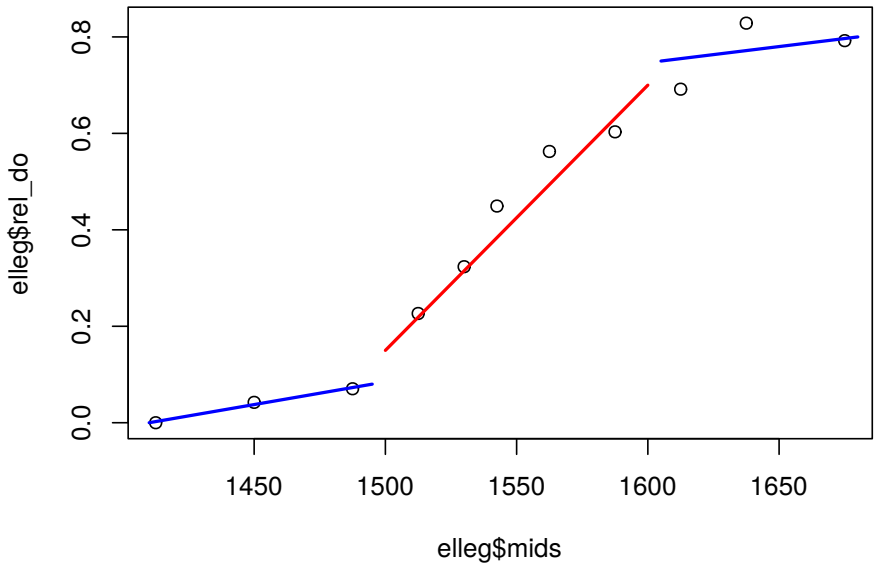
- We’re going to focus on (3) for now, and revisit (1) and (2) later



The time dimension

- In principle we could try and model linguistic changes using three line segments
- But this is both
  1. technically difficult (where do we decide one segment ends and the other begins?)
  2. *a priori* unattractive (*natura non facit saltus*)

- Insight: the change looks like an elongated 'S'
- We can look for ways of describing such **S-curves**



Logistic functions

- The **STANDARD LOGISTIC FUNCTION** is the function

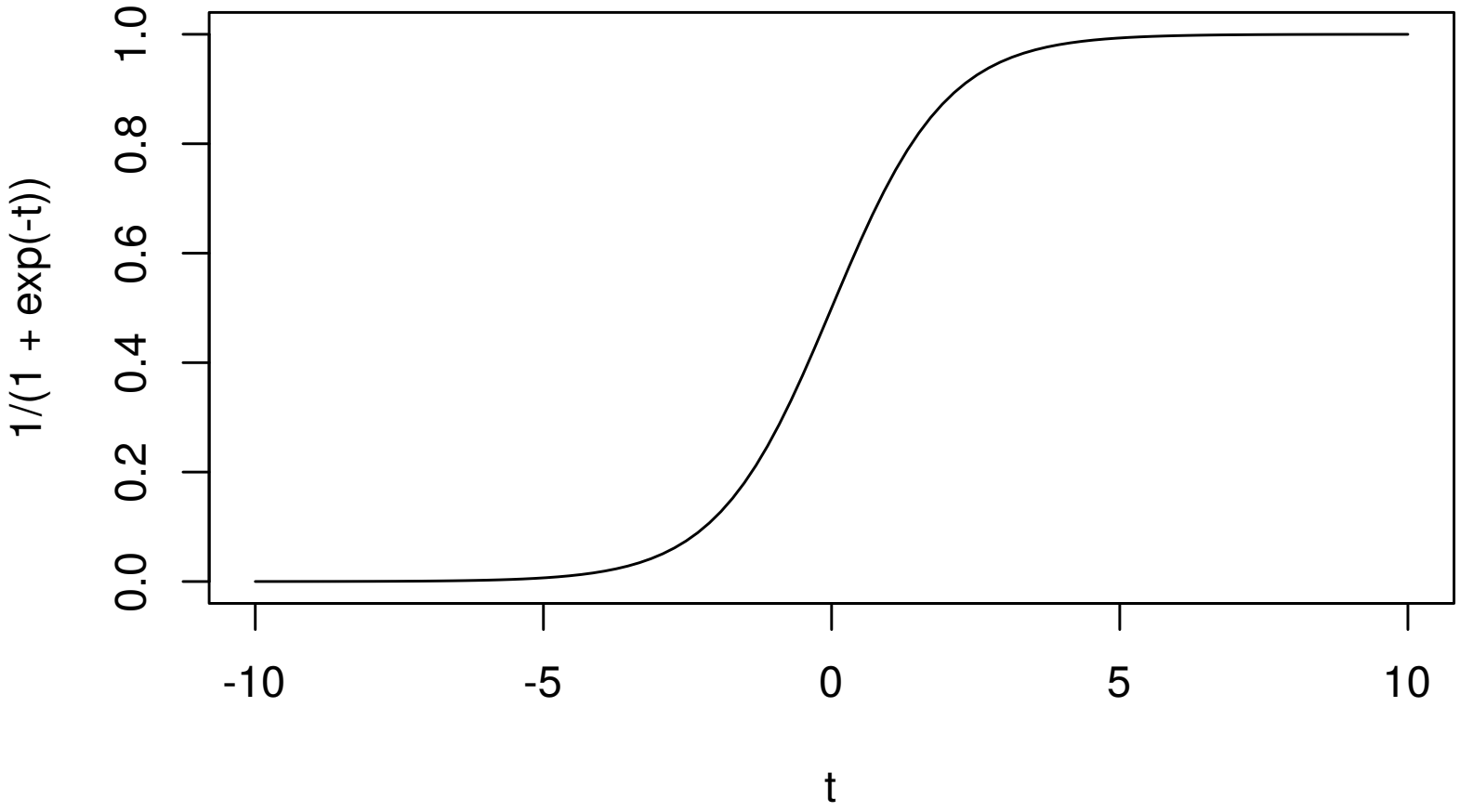
$$f(t) = \frac{1}{1 + \exp(-t)}$$

- Here
  - t is time
  - $f(t)$ is the relative frequency at t
  - exp is the **NATURAL EXPONENTIAL FUNCTION**: $\exp(x) = e^x$, where e is Euler's number ($e \approx 2.718$)

```
t <- seq(from=-10, to=10, length.out=100)
plot(t, 1/(1 + exp(-t)), type="l")
```

Logistic functions

```r
t <- seq(from=-10, to=10, length.out=100)
plot(t, 1/(1 + exp(-t)), type="l")
```



Logistic functions

- That's nice, but the curve is always in the same place on the time axis
- Let's add a **PARAMETER**, k:

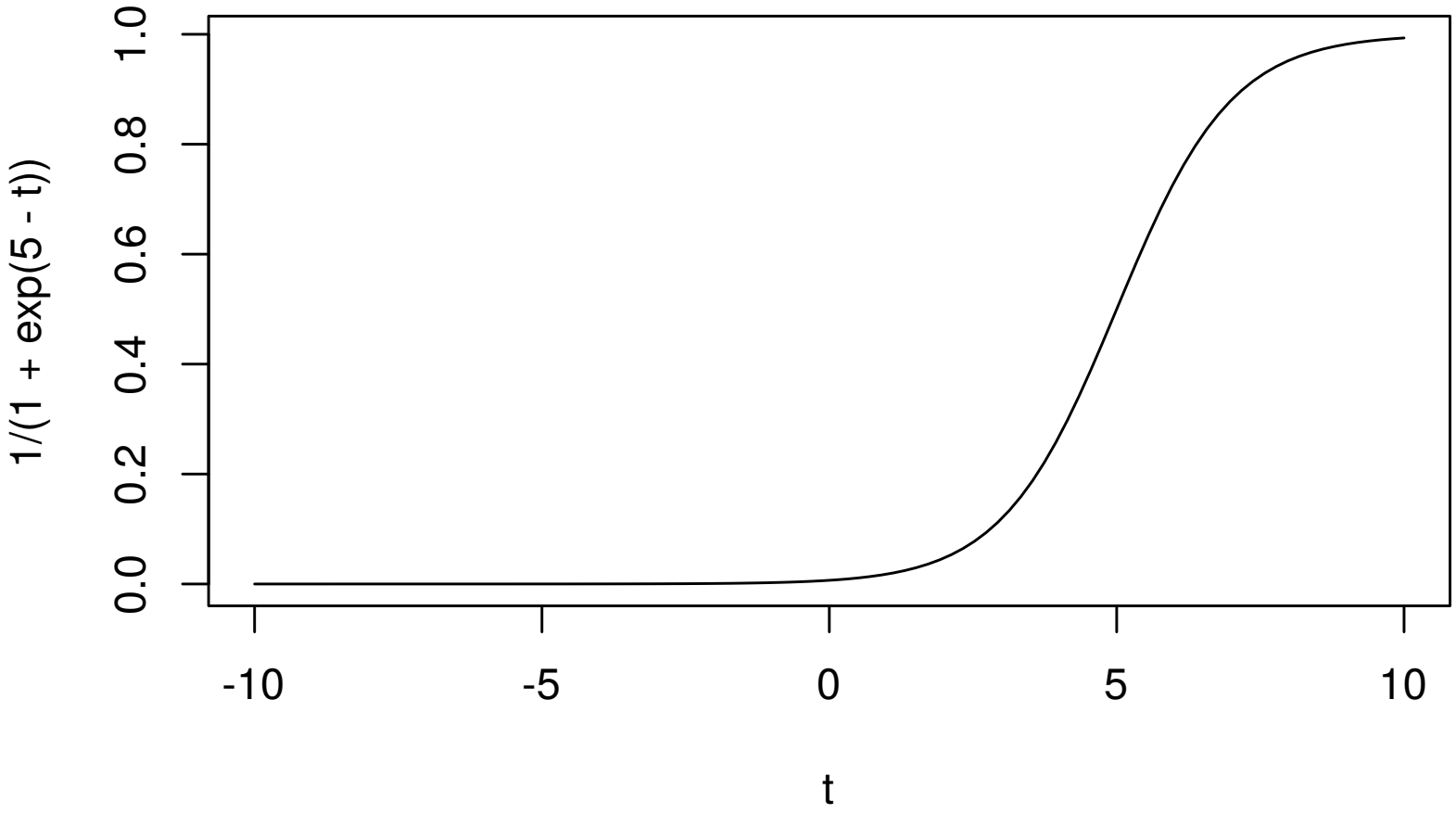
$$f(t) = \frac{1}{1 + \exp(k - t)}$$

- A parameter has a preset value (unlike f, it does not vary as a function of time t)
- Try:

```
plot(t, 1/(1 + exp(5 - t)), type="l")
plot(t, 1/(1 + exp(-5 - t)), type="l")
plot(t, 1/(1 + exp(0 - t)), type="l")
```

Logistic functions

```
t <- seq(from=-10, to=10, length.out=100)
plot(t, 1/(1 + exp(5 - t)), type="l")
```



Logistic functions

- That's nice, but the curve always has the same steepness
- Let's add another parameter, s:

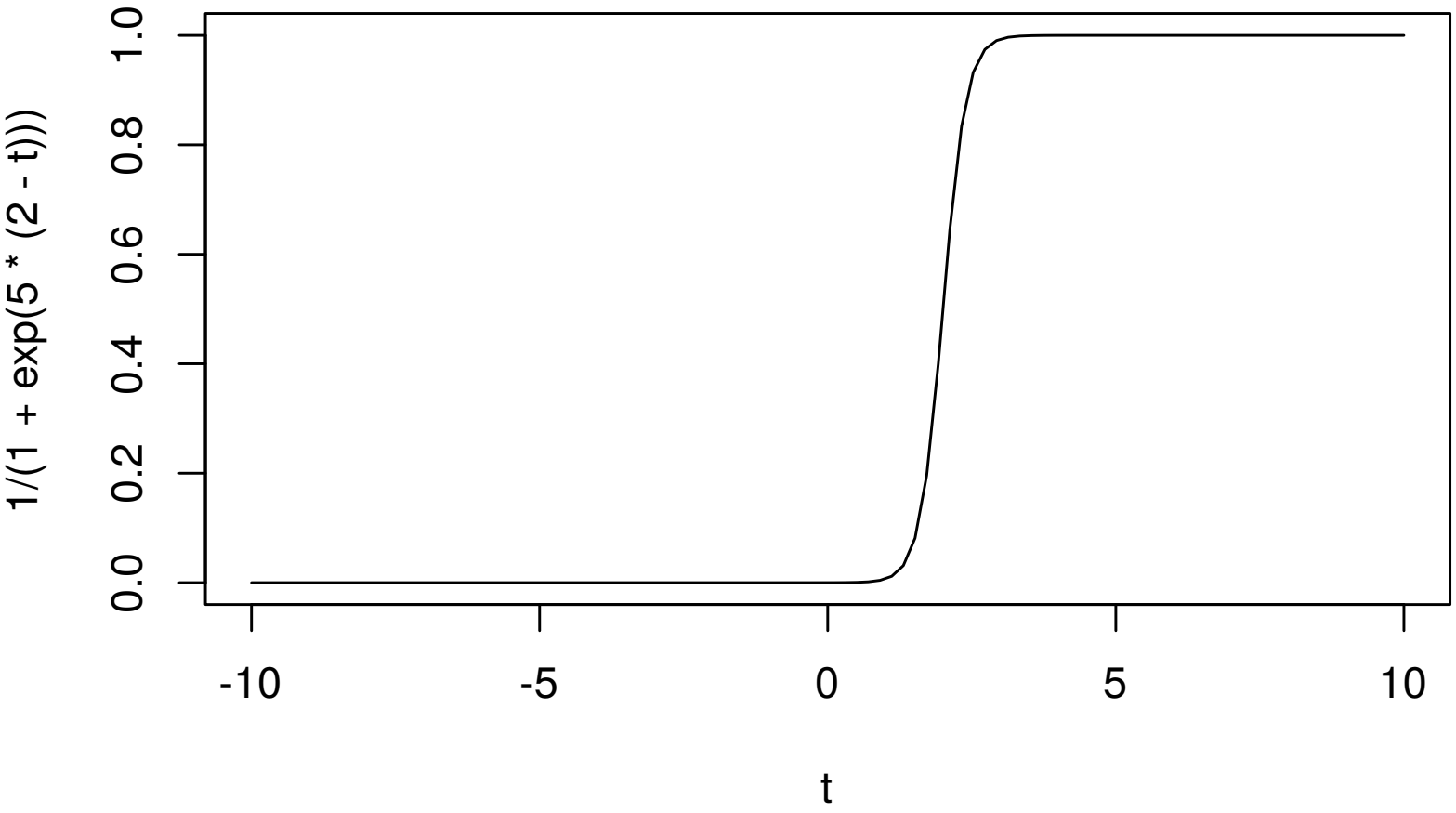
$$f(t) = \frac{1}{1 + \exp(s \times (k - t))}$$

- Try:

```
plot(t, 1/(1 + exp(5*(2 - t))), type="l")
plot(t, 1/(1 + exp(0.5*(2 - t))), type="l")
plot(t, 1/(1 + exp(-0.5*(2 - t))), type="l")
```

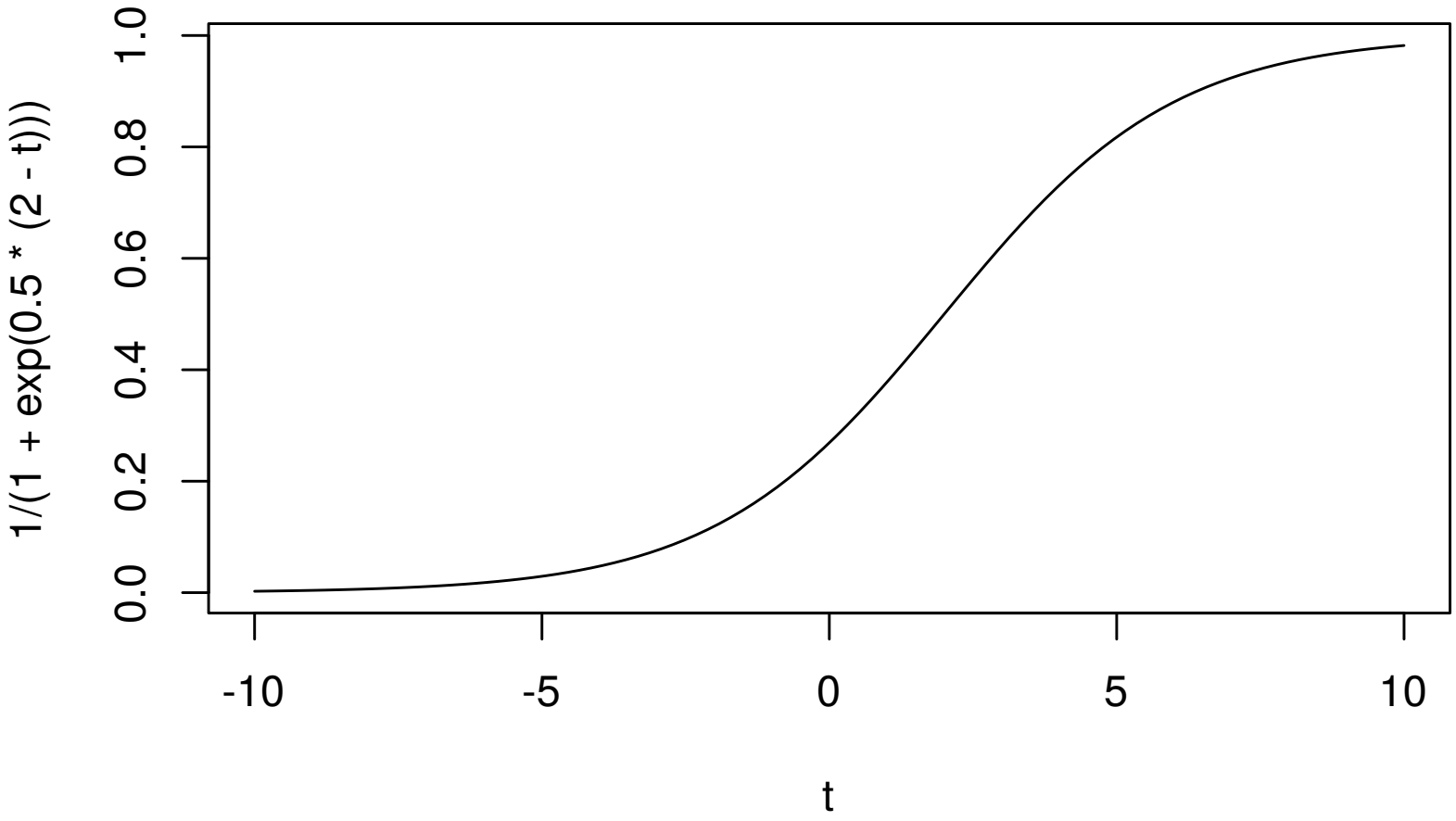
Logistic functions

```
plot(t, 1/(1 + exp(5*(2 - t))), type="l")
```



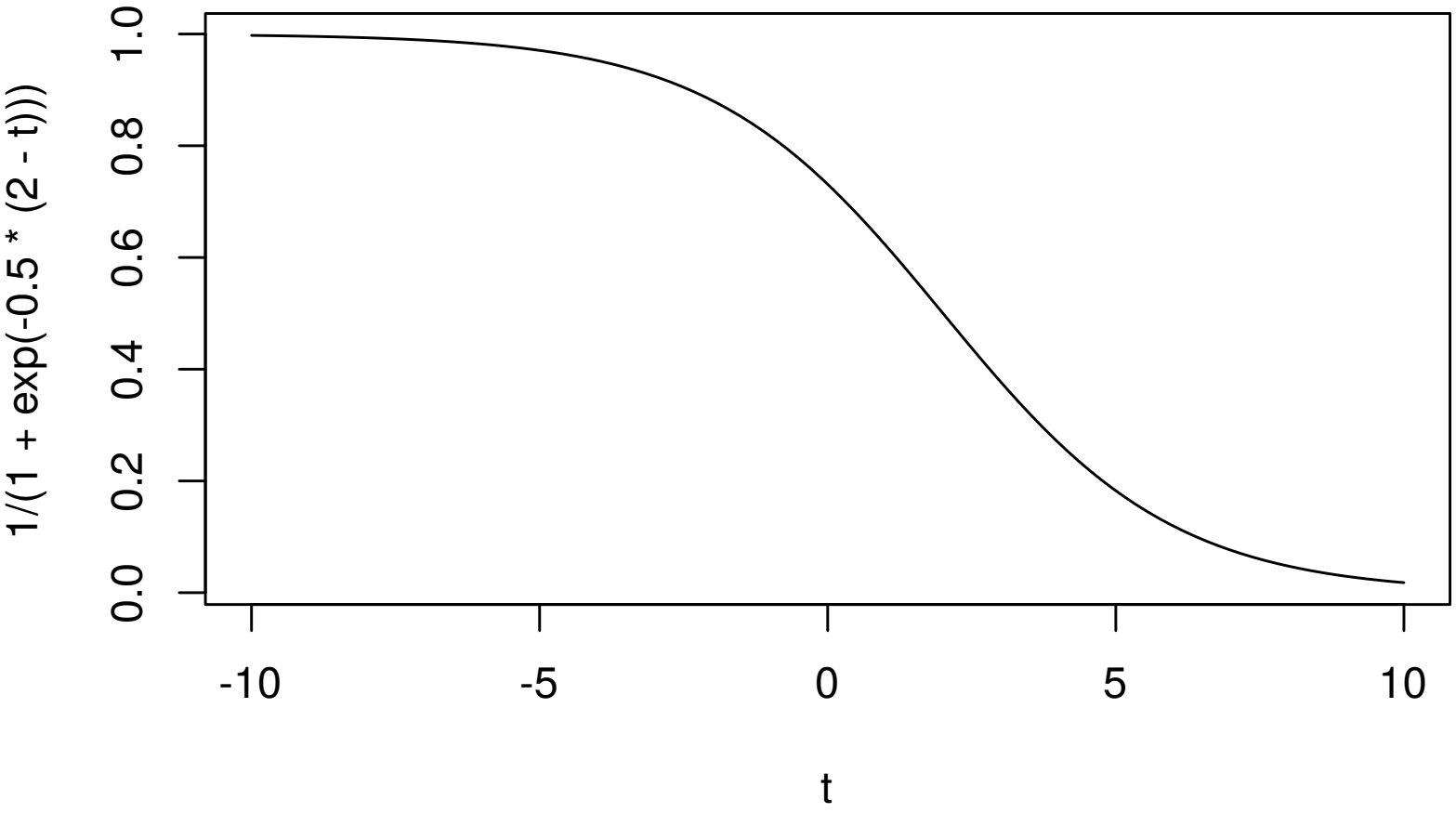
Logistic functions

```r
plot(t, 1/(1 + exp(0.5*(2 - t))), type="l")
```



Logistic functions

```r
plot(t, 1/(1 + exp(-0.5*(2 - t))), type="l")
```



Logistic functions

- With s and k, we have defined a **FAMILY** of S-curves
- The parameter k controls the…

Logistic functions

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- The parameter k controls the… **time position of the change**
  - For positive k, the change is shifted…

Logistic functions

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- The parameter k controls the…**time position of the change**
  - For positive k, the change is shifted…**to the right**
  - For negative k, the change is shifted…

Logistic functions

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- The parameter s controls the…

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- The parameter s controls the…**steepness of the change**
  - For positive s, the relative frequency…

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  - For positive s, the relative frequency… **increases over time**
  - For negative s, the relative frequency…

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  - For positive s, the relative frequency…**increases over time**
  - For negative s, the relative frequency…**decreases over time**
  - The larger the absolute value $|s|$, the…**steeper the change**
- s is known as the **SLOPE**, k the **INTERCEPT**
- We can now (next week) try to precisely describe patterns in historical data, by finding out which combination of s and k gives the best fit!

New Portfolio Exercise etc.

- On ILIAS, you will now find a new Portfolio Exercise on S-curves
- Next week we will continue with this same theme
- I am also in the process of preparing a comprehensive tutorial on plots/figures in R – it will be available next week